

Software Requirements Specification (SRS)

Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform

PO2 Assessment Tool 1

Scenario-Based Requirement Analysis

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1. INTRODUCTION

1.1 PURPOSE

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for an Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform. It is intended to serve as the primary reference for the design, development, testing, and validation of the proposed system, and as an agreement between stakeholders and the development team on the scope and behavior of the system.

1.2 INTENDED AUDIENCE

This document is intended for warehouse operations management, IT/solution architects, software developers, quality assurance teams, project sponsors, and third-party integration partners (ERP, RFID/IoT hardware, and AMR vendors) involved in the design and delivery of the platform.

1.3 BUSINESS DOMAIN

The system operates within the logistics and supply chain management domain, specifically warehouse management systems (WMS) enhanced with Artificial Intelligence, RFID, Internet of Things (IoT), Autonomous Mobile Robots (AMRs), and cloud computing to automate storage, retrieval, and inventory operations.

1.4 REFERENCES

- Problem Statement: "Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform" (Assignment Brief 12).
- IEEE 830-1998 / ISO/IEC/IEEE 29148:2018 - Software Requirements Specification guidelines.
- Industry references on RFID/IoT-enabled warehouse management and AMR-based fulfillment operations.

1.5 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Definition
AMR	Autonomous Mobile Robot
RFID	Radio Frequency Identification
IoT	Internet of Things
WMS	Warehouse Management System
ERP	Enterprise Resource Planning
SKU	Stock Keeping Unit
API	Application Programming Interface
SRS	Software Requirements Specification

Term	Definition
KPI	Key Performance Indicator
RTO / RPO	Recovery Time Objective / Recovery Point Objective

2. PROBLEM DESCRIPTION

Warehouses are increasingly reliant on manual or semi-automated processes for inventory tracking, storage allocation, and order fulfillment. These manual operations are prone to human error, resulting in inventory discrepancies, inefficient use of storage space, and delayed order processing. Existing warehouse systems typically provide limited real-time visibility into inventory movement, making it difficult for warehouse managers to make timely, data-driven decisions.

The proposed Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform addresses these challenges by combining RFID and IoT sensing technologies for automatic, real-time inventory tracking; AI-driven algorithms for optimal storage allocation; and integration with Autonomous Mobile Robots and existing ERP/logistics systems to streamline order fulfillment. The platform is designed to improve inventory accuracy, warehouse productivity, and overall operational efficiency while reducing costs associated with manual errors and inefficient space utilization.

2.1 STAKEHOLDERS

- Warehouse Managers - responsible for overall warehouse performance and decision-making.
- Inventory Operators / Warehouse Staff - perform day-to-day receiving, put-away, picking, and packing tasks.
- System Administrators / IT Team - configure, maintain, and secure the platform and its integrations.
- Customers / Order Clients - place orders and expect timely, accurate fulfillment and status visibility.
- Business/Project Sponsors - fund the project and expect measurable ROI (cost reduction, productivity gains).
- Technology Partners - RFID/IoT hardware vendors, AMR vendors, ERP/logistics software providers.

2.2 EXISTING CHALLENGES

- Manual inventory counts result in frequent discrepancies between recorded and actual stock.
- Inefficient storage allocation leads to wasted space and longer retrieval times.
- Lack of real-time visibility delays decision-making and slows order fulfillment.
- Disconnected systems (inventory, order management, logistics) cause data silos and synchronization errors.
- Limited reporting capability hampers performance monitoring and continuous improvement.

3. SCOPE OF THE SYSTEM

The system will provide end-to-end automation of warehouse inventory tracking and order fulfillment support, including:

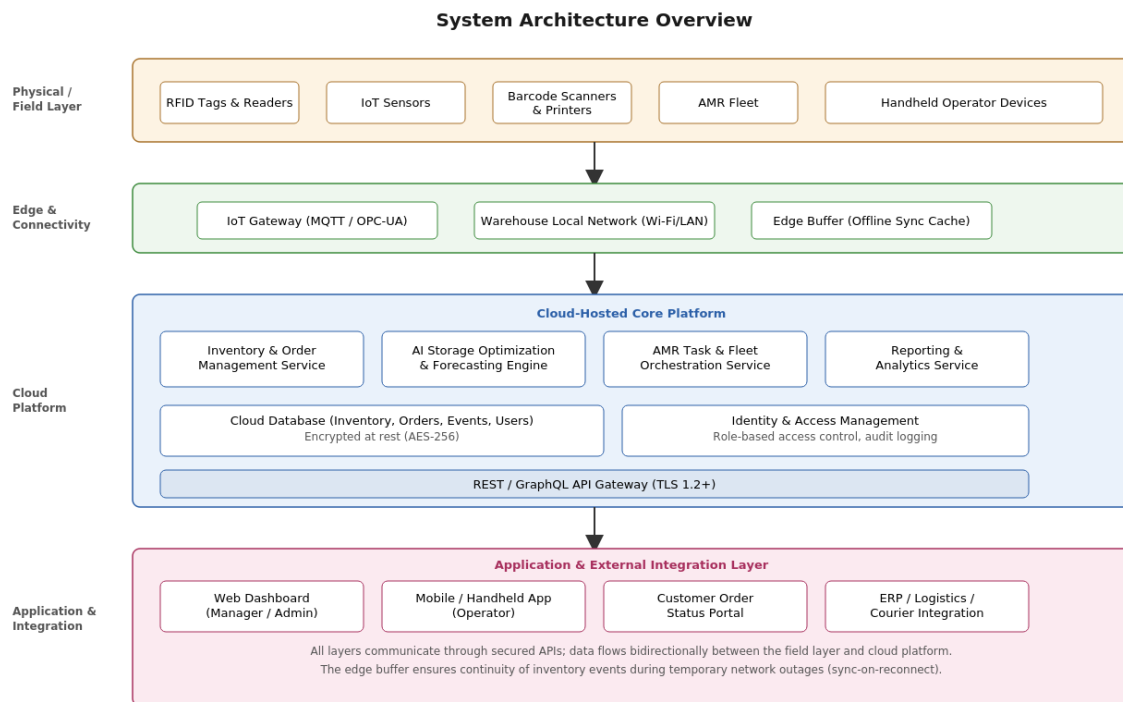
- RFID/IoT-based automated inventory identification, location tracking, and condition monitoring (e.g., temperature/humidity where applicable).
- AI-based storage location optimization and dynamic re-slotting recommendations.
- Real-time dashboards for inventory visibility, AMR task tracking, and warehouse KPIs.
- Order fulfillment workflow support including pick-list generation, task assignment, and status synchronization with ERP/logistics systems.

- Discrepancy detection and reconciliation through automated cycle counts and audits.
- Configurable performance and analytics reporting.
- Role-based user and system administration.

Out of Scope (Phase 1): Full autonomous robotic picking (physical item manipulation) is limited to AMR-assisted transport; direct integration with financial/accounting systems beyond inventory valuation feeds; and predictive demand forecasting beyond baseline analytics (may be considered in a later phase).

3.1 SYSTEM ARCHITECTURE OVERVIEW

The platform follows a layered architecture, from field-level sensing and robotics hardware, through edge connectivity, to a cloud-hosted core platform and integrated applications. This layering isolates hardware dependencies from business logic and allows each layer to scale and be maintained independently.



4. FUNCTIONAL REQUIREMENTS

Functional requirements are grouped by module below. Each requirement is uniquely identified and assigned a priority (High/Medium/Low) to support release planning.

A. INVENTORY TRACKING & RFID/IOT INTEGRATION

ID	Requirement Description	Priority
FR-01	The system shall capture and register inventory item data via RFID tags and barcode scans at receiving docks.	High
FR-02	The system shall read RFID/IoT sensor signals in real time to detect item location within the warehouse.	High
FR-03	The system shall automatically update inventory records when items are moved, picked, packed, or shipped.	High

ID	Requirement Description	Priority
FR-04	The system shall maintain a complete audit trail (timestamp, user/device, location) for every inventory transaction.	Medium
FR-05	The system shall support bulk inventory registration through file import (CSV/Excel) and API integration.	Medium

B. AI-BASED STORAGE OPTIMIZATION

ID	Requirement Description	Priority
FR-06	The system shall use AI/ML algorithms to recommend optimal storage locations based on item size, turnover rate, and demand patterns.	High
FR-07	The system shall dynamically re-slot fast-moving items closer to dispatch areas to minimize retrieval time.	Medium
FR-08	The system shall predict storage space requirements using historical inventory and seasonal demand data.	Medium
FR-09	The system shall recommend put-away locations to warehouse operators at the time of goods receipt.	High

C. REAL-TIME MONITORING & VISIBILITY

ID	Requirement Description	Priority
FR-10	The system shall display a real-time dashboard of inventory levels, item locations, and warehouse activity.	High
FR-11	The system shall track AMR (Autonomous Mobile Robot) location, status, and task progress in real time.	Medium
FR-12	The system shall generate real-time alerts for low stock, misplaced items, equipment faults, or abnormal temperature/humidity readings.	High
FR-13	The system shall provide a searchable interface to locate any item by SKU, batch, or RFID tag ID within seconds.	High

D. ORDER FULFILLMENT & LOGISTICS

ID	Requirement Description	Priority
FR-14	The system shall generate optimized pick lists and pick paths for warehouse staff and AMRs based on order priority.	High
FR-15	The system shall automatically update order status (received, picked, packed, shipped, delivered) and notify the customer/ERP.	High
FR-16	The system shall integrate with external ERP, WMS, and logistics/courier systems via APIs for order and shipment synchronization.	High
FR-17	The system shall support batch and wave picking strategies to improve order throughput.	Medium

E. INVENTORY ACCURACY & DISCREPANCY MANAGEMENT

ID	Requirement Description	Priority
FR-18	The system shall automatically compare RFID/IoT scan data against system records and flag discrepancies.	High
FR-19	The system shall support scheduled and ad-hoc cycle counting with mobile-device-assisted audits.	Medium
FR-20	The system shall allow authorized staff to review, approve, and reconcile flagged discrepancies with reason codes.	Medium

F. REPORTING & ANALYTICS

ID	Requirement Description	Priority
FR-21	The system shall generate configurable reports on inventory accuracy, order fulfillment time, storage utilization, and labor productivity.	High
FR-22	The system shall provide exportable reports in PDF/Excel formats and support scheduled email delivery.	Medium
FR-23	The system shall provide predictive analytics (e.g., demand forecasting, stock-out risk) using historical data.	Low

G. USER & SYSTEM ADMINISTRATION

ID	Requirement Description	Priority
FR-24	The system shall support role-based access control (Administrator, Warehouse Manager, Inventory Operator, Auditor).	High
FR-25	The system shall allow administrators to configure warehouse zones, racks, bins, and device (RFID/IoT/AMR) registrations.	Medium
FR-26	The system shall log all user actions for security auditing and compliance purposes.	Medium

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes the system must satisfy in addition to its functional behavior.

Category	Requirement
Performance	The system shall process RFID/IoT scan events and update inventory records within 2 seconds under normal load; dashboards shall refresh within 3 seconds.
Scalability	The system shall support scaling from a single warehouse to a multi-warehouse network handling at least 1,000,000 SKUs and 5,000 concurrent sensor/device connections without redesign.
Reliability	The system shall achieve 99.5% uptime for core tracking and order-processing services and shall not lose inventory transaction data during transient network failure (offline buffering with sync-on-reconnect).

Category	Requirement
Availability	Core warehouse operations (tracking, picking, dispatch) shall be available 24x7, excluding planned maintenance windows communicated at least 48 hours in advance.
Security	The system shall enforce role-based access control, encrypt data in transit (TLS 1.2+) and at rest (AES-256), and comply with applicable data protection regulations.
Usability	Warehouse-floor interfaces (handheld/mobile) shall be operable with minimal training (under 2 hours) and support barcode/RFID scan-first workflows with large touch targets.
Maintainability	The system shall use a modular, service-oriented architecture allowing independent updates to AI, IoT integration, and reporting modules without full-system downtime.
Interoperability	The system shall expose REST/GraphQL APIs and support standard protocols (MQTT, OPC-UA) for integration with ERP, AMR fleets, and RFID/IoT hardware from multiple vendors.
Auditability	All inventory-affecting transactions shall be logged with user/device ID, timestamp, and before/after values, retained for a minimum of 3 years.
Portability	The web dashboard shall function correctly on major browsers (Chrome, Edge, Firefox, Safari); mobile apps shall support Android 10+ and iOS 14+.
Disaster Recovery	The system shall support automated daily backups with a Recovery Point Objective (RPO) of 1 hour and Recovery Time Objective (RTO) of 4 hours.

6. ACTORS AND USE CASES

6.1 ACTORS

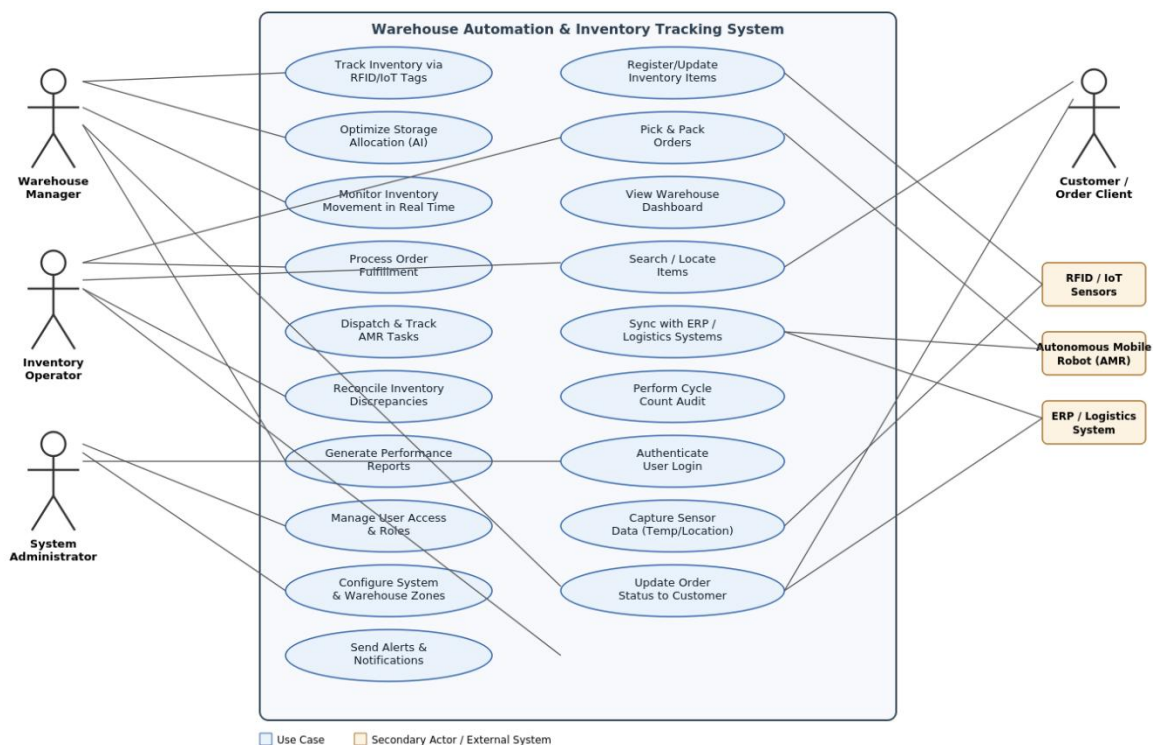
Actor	Type	Description
Warehouse Manager	Primary	Oversees warehouse operations; views dashboards, approves storage optimization plans, monitors KPIs, reviews performance reports, resolves escalated discrepancies.
Inventory Operator / Warehouse Staff	Primary	Performs receiving, put-away, picking, packing, and cycle counting using handheld RFID/barcode devices; updates inventory transactions.
System Administrator	Primary	Configures system settings, warehouse zones/bins, user roles and permissions, integrates and registers RFID/IoT devices and AMRs.
Customer / Order Client	Primary	Places orders (via e-commerce/ERP front-end) and receives order status/tracking updates generated by the system.

Actor	Type	Description
RFID/IoT Sensors & Readers	Secondary (System)	Automatically capture tag reads, location, temperature, and humidity data and feed it into the platform.
Autonomous Mobile Robots (AMR)	Secondary (System)	Execute automated pick, transport, and put-away tasks assigned by the system; report task status and location.
ERP / Logistics / Courier Systems	Secondary (System)	Exchange order, shipment, and inventory data with the platform via APIs for end-to-end supply chain visibility.

6.2 USE CASE DIAGRAM

The diagram below illustrates the primary and secondary actors and their interactions with the Warehouse Automation and Inventory Tracking System.

Use Case Diagram - Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform



6.3 USE CASE DESCRIPTIONS

Detailed specifications for the major use cases are provided below.

UC-01: Track Inventory via RFID/IoT

Use Case ID	UC-01
Use Case Name	Track Inventory via RFID/IoT
Actors	Inventory Operator, RFID/IoT Sensors
Description	Automatically capture and update the location and status of inventory items as they move through the warehouse.
Preconditions	RFID tags are attached to items/pallets; readers are installed and operational.

Main Flow	1. RFID/IoT sensor detects a tag read event as an item passes a reader zone. 2. System receives the sensor event and identifies the item and its new location. 3. System updates the inventory record and location history in real time. 4. System reflects the updated location on the live warehouse dashboard.
Alternate Flow	If the tag read is invalid/duplicate, the system discards the event and logs an error for review.
Postconditions	Inventory location and movement history are current and accurate.

UC-02: Optimize Storage Allocation (AI)

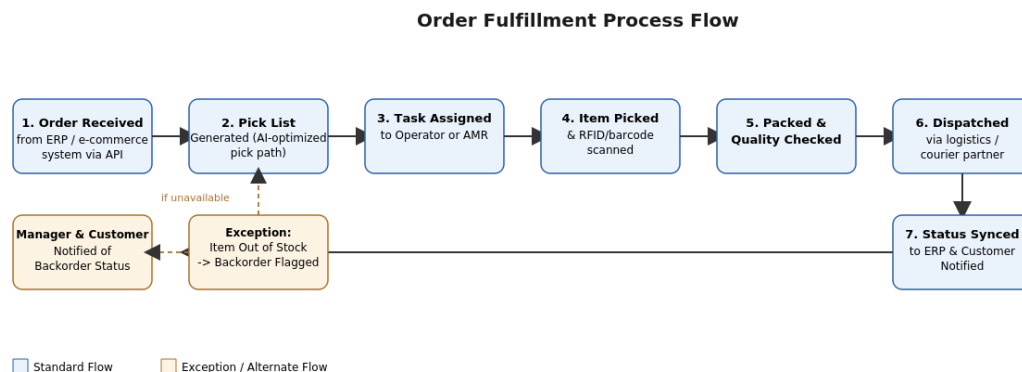
Use Case ID	UC-02
Use Case Name	Optimize Storage Allocation (AI)
Actors	Warehouse Manager, System (AI Engine)
Description	Recommend optimal storage/put-away locations for incoming or repositioned inventory using AI-driven analysis of turnover, size, and demand.
Preconditions	Historical inventory movement and item master data are available to the AI engine.
Main Flow	1. Operator scans a newly received item for put-away. 2. System's AI engine analyzes item attributes, turnover rate, and current warehouse occupancy. 3. System recommends an optimal bin/rack location to the operator's handheld device. 4. Operator confirms placement; system updates the storage map.
Alternate Flow	If the recommended location is unavailable, the system suggests the next-best alternative location.
Postconditions	Item is stored in an AI-optimized location; storage map is updated.

UC-03: Process Order Fulfillment

Use Case ID	UC-03
Use Case Name	Process Order Fulfillment
Actors	Inventory Operator, Customer, ERP System
Description	Generate optimized pick lists, track picking/packing/shipping progress, and notify stakeholders of order status.
Preconditions	A valid sales order has been received from the ERP/e-commerce system.
Main Flow	1. System receives order details from the ERP system via API. 2. System generates an optimized pick list/path and assigns it to an operator or AMR.

	3. Operator/AMR picks and packs items; each action is scanned and recorded. 4. System updates order status and notifies the customer and ERP system of dispatch.
Alternate Flow	If an item is out of stock, the system flags a backorder and notifies the manager and customer.
Postconditions	Order is fulfilled and shipped; status is synchronized across systems.

The diagram below traces the end-to-end order fulfillment process referenced in UC-03, including the exception path for out-of-stock items.



UC-04: Reconcile Inventory Discrepancies

Use Case ID	UC-04
Use Case Name	Reconcile Inventory Discrepancies
Actors	Inventory Operator, Warehouse Manager
Description	Detect, review, and resolve mismatches between physical (RFID-scanned) inventory and system records.
Preconditions	Cycle count or automated scan data is available for comparison against system records.
Main Flow	1. System compares scanned inventory counts with recorded quantities. 2. System flags SKUs/locations with discrepancies above a configurable threshold. 3. Operator investigates and records findings (e.g., miscount, damage, theft). 4. Manager reviews and approves the adjustment; system updates inventory records.
Alternate Flow	If discrepancy exceeds a critical threshold, system escalates an alert to the manager immediately.
Postconditions	Inventory records reflect verified, accurate quantities.

UC-05: Generate Warehouse Performance Reports

Use Case ID	UC-05
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Use Case Name	Generate Warehouse Performance Reports
Actors	Warehouse Manager, System Administrator
Description	Produce configurable reports on inventory accuracy, fulfillment time, storage utilization, and productivity.
Preconditions	Sufficient transactional data exists for the selected reporting period.
Main Flow	1. Manager selects report type, date range, and filters from the reporting module. 2. System aggregates relevant transaction and performance data. 3. System generates the report and displays it on-screen with visual charts. 4. Manager exports the report (PDF/Excel) or schedules recurring delivery.
Alternate Flow	If no data is available for the selected filters, the system displays an appropriate message.
Postconditions	Report is generated, viewable, and available for export/distribution.

UC-06: Dispatch & Track AMR Tasks

Use Case ID	UC-06
Use Case Name	Dispatch & Track AMR Tasks
Actors	System Administrator, Autonomous Mobile Robot
Description	Assign transport/picking tasks to AMRs and monitor their execution in real time.
Preconditions	AMR fleet is registered, charged, and connected to the system network.
Main Flow	1. System generates a transport/pick task based on order or replenishment needs. 2. System assigns the task to an available, optimally positioned AMR. 3. AMR executes the task and streams status/location updates to the system. 4. System marks the task complete and updates related inventory/order records.
Alternate Flow	If the assigned AMR becomes unavailable (low battery/obstacle), system reassigns the task to another unit.
Postconditions	Task is completed and reflected in inventory/order status; AMR is available for the next task.

7. DATA REQUIREMENTS

The system shall manage the following core data entities. A relational or document-based cloud data store shall persist this data with appropriate indexing for real-time query performance.

Data Entity	Key Attributes
Item Master	SKU ID, Item Name, Category, Dimensions, Weight, Barcode/RFID Tag ID, Unit of Measure, Shelf-life
Inventory Record	Item SKU, Warehouse ID, Zone/Rack/Bin ID, Quantity On-Hand, Reserved Quantity, Last Updated Timestamp
Location/Bin Master	Location ID, Zone, Aisle, Rack, Bin Capacity, Bin Status, Storage Constraints
RFID/IoT Event Log	Event ID, Tag ID, Reader ID, Timestamp, Location Coordinates, Sensor Reading (temp/humidity)
Order	Order ID, Customer ID, Order Date, Line Items, Priority, Status, Required Ship Date
Pick/Pack Task	Task ID, Order ID, Assigned Operator/AMR, Pick Path, Status, Start/End Timestamp
AMR Fleet Record	Robot ID, Current Location, Battery Level, Task Queue, Operational Status
Discrepancy Record	Discrepancy ID, SKU, Location, Expected Qty, Actual Qty, Reason Code, Resolution Status
User Account	User ID, Name, Role, Permissions, Login History
Performance Report	Report ID, Type, Date Range, Generated Metrics, Export Format

Data retention: transactional data (inventory events, order history) shall be retained for a minimum of 3 years to support auditing and analytics, subject to applicable data protection regulations.

8. EXTERNAL INTERFACE REQUIREMENTS

8.1 USER INTERFACES

- Web-based dashboard for warehouse managers and administrators (desktop/tablet responsive).
- Mobile/handheld application for inventory operators supporting barcode/RFID scan-first workflows.
- Customer-facing order status interface (web/mobile), typically embedded within the existing e-commerce or ERP front-end.

8.2 HARDWARE INTERFACES

- RFID readers and tags (fixed portal readers at dock doors and handheld readers).
- IoT sensors for environmental monitoring (temperature, humidity) where applicable to stored goods.
- Autonomous Mobile Robots (AMR) with onboard connectivity for task and telemetry data exchange.
- Barcode scanners and label printers at receiving/packing stations.

8.3 SOFTWARE INTERFACES

- REST/GraphQL APIs for integration with ERP and logistics/courier systems.
- MQTT/OPC-UA protocol support for IoT device and AMR communication.
- Cloud infrastructure services (compute, storage, and managed database) for hosting the platform and AI models.

8.4 COMMUNICATION INTERFACES

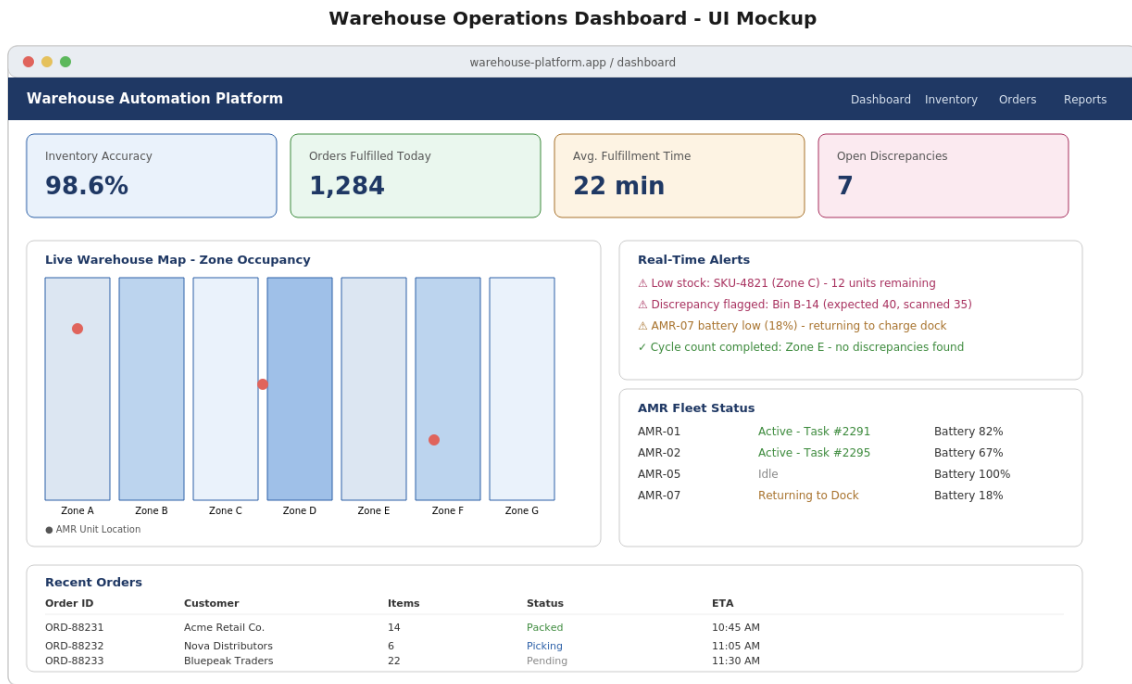
- Secure HTTPS/TLS for all web and API communication.
- Wi-Fi/LAN network connectivity throughout the warehouse for real-time device communication.
- Notification interfaces (email/SMS/push) for alerts to managers and customers.

9. SYSTEM FEATURES

Feature	Priority	Description
Real-Time Inventory Tracking	High	Automatic capture and continuous update of item location and status using RFID/IoT, eliminating manual stock checks.
AI Storage Optimization Engine	High	Machine-learning-driven recommendations for put-away and re-slotting to maximize space utilization and minimize travel time.
Live Operations Dashboard	High	Consolidated real-time view of inventory levels, order status, AMR activity, and alerts for decision-makers.
Automated Order Fulfillment Workflow	High	End-to-end digital workflow from order receipt to pick, pack, and dispatch, synchronized with ERP/logistics systems.
Discrepancy Detection & Reconciliation	Medium	Automated comparison of physical scans to system records with guided reconciliation workflow.
Configurable Reporting & Analytics	Medium	Customizable reports on accuracy, productivity, and utilization with export and scheduling options.
AMR Fleet Integration	Medium	Task assignment, routing, and real-time monitoring of autonomous mobile robots.
Role-Based Access & Administration	High	Secure, configurable user roles and permissions with full audit logging.

9.1 ILLUSTRATIVE DASHBOARD MOCKUP

The mockup below illustrates the primary manager/administrator dashboard, combining KPIs, a live warehouse zone-occupancy map with AMR positions, real-time alerts, AMR fleet status, and a recent-orders view - consolidating several of the features above into a single operational screen.



10. ASSUMPTIONS AND CONSTRAINTS

10.1 ASSUMPTIONS

- All inventory items intended for automated tracking will be tagged with RFID tags or barcodes prior to entering the warehouse.
- The warehouse facility has reliable network coverage (Wi-Fi/LAN) throughout all zones to support real-time IoT and AMR communication.
- RFID/IoT hardware and AMR units used are compatible with standard communication protocols (MQTT, REST, OPC-UA).
- Warehouse staff will receive basic training on handheld scanning devices and the operator interface.
- Historical inventory and order data will be available for AI model training and storage-optimization recommendations.
- ERP and logistics/courier partner systems expose APIs suitable for integration.
- The organization will provide the necessary power, network infrastructure, and physical space for AMR charging stations and IoT gateways.

10.2 CONSTRAINTS

Type	Constraint
Technical	The system must integrate with existing legacy ERP/WMS systems that may have limited or outdated APIs, requiring middleware or custom adapters.
Technical	AI storage-optimization accuracy depends on the volume and quality of historical data available at go-live; early recommendations may be less optimal.
Operational	Warehouse operations cannot be halted for extended periods during rollout; phased deployment and parallel-run with existing manual processes is required.

Type	Constraint
Operational	Staff availability for training must be scheduled around shift patterns without disrupting order fulfillment SLAs.
Economic	Budget constraints may limit the initial scale of RFID tag deployment and the number of AMR units procured in Phase 1.
Economic	Total cost of ownership includes recurring costs for IoT connectivity, cloud hosting, and hardware maintenance.
Legal/Regulatory	The system must comply with data protection regulations (e.g., applicable data privacy laws) for any customer or employee data processed.
Legal/Regulatory	RFID/IoT devices must comply with regional wireless spectrum and safety certification requirements.
Time	The project must be delivered within the sponsor-defined timeline; phased delivery (tracking module first, AI optimization and AMR integration in later phases) may be required to meet milestones.

11. VALIDATION OF REQUIREMENT COMPLETENESS AND CONSISTENCY

The requirements captured in this SRS were reviewed against the original problem statement's objectives and expected outcomes to verify coverage, and were checked for common quality issues as summarized below.

Check	Findings / Approach
Completeness	Cross-checked each objective (1-8) in the problem statement against functional requirements; all objectives map to at least one FR. No stakeholder-stated need was found unaddressed.
Consistency	Verified that NFR performance targets (e.g., 2-second update latency) do not conflict with reliability requirements (offline buffering); reconciled by defining buffered updates as compliant once connectivity resumes.
Ambiguity	Reviewed requirements for vague terms (e.g., 'fast', 'user-friendly'); replaced with measurable criteria (e.g., specific time thresholds, training-time limits) where identified.
Redundancy	Checked for overlapping requirements between inventory tracking (FR-01 to FR-05) and discrepancy management (FR-18 to FR-20); confirmed no duplication as their triggers and outcomes differ.
Traceability	Each functional requirement is traceable to a use case and, in turn, to a stated business objective or expected outcome, supporting future test-case derivation.
Testability	Reviewed NFRs to ensure each includes a quantifiable threshold (percentages, time limits, counts) so they can be objectively verified during acceptance testing.

11.1 SUGGESTED IMPROVEMENTS

- Conduct structured stakeholder workshops (warehouse operations, IT, finance, and customer service) to validate assumptions before development begins.

- Develop a formal Requirements Traceability Matrix (RTM) linking business objectives, functional requirements, use cases, and test cases.
- Define measurable acceptance criteria for each NFR (already partially addressed) and review them with operations stakeholders for real-world feasibility.
- Prototype the AI storage-optimization module early using historical data to validate assumptions about data availability and model accuracy.
- Introduce a formal change-management process to handle scope changes arising from phased AMR/IoT hardware procurement constraints.

Conduct a requirements walkthrough/inspection with an independent reviewer to catch ambiguity or missing edge cases (e.g., damaged tags, network outages, multi-warehouse conflicts)